

SHETH L.U.J. AND SIR M.V. DEGREE COLLEGE OF SCIENCE

SUBJECT NAME: S101 Data Analysis with SAS / SPSS / R Practical

PRACTICAL - 11 (M2)

AIM: Generating histograms and box plots using ggplot2 (R).

Input:

```
# S101 RAJDEEPP .M. PARAB
# Data Analysis with SAS / SPSS / R Practical
# PRACTICAL 11 - Generating histograms and box plots using ggplot2 (R).
# Histograms and Boxplots
library(ggplot2)
library(dplyr)
# Load dataset
df <- read.csv("retail_sales_dataset.csv")

# 1. choose numeric and categorical columns
num_cols <- names(df)[sapply(df, is.numeric)]
cat_cols <- names(df)[sapply(df, is.character) | sapply(df, is.factor)]
num_var <- num_cols[1] # numeric variable
cat_var <- cat_cols[1] # categorical variable
cat("\nNumeric used:", num_var)
cat("\nCategorical used:", cat_var, "\n")

# 2. take a small clean subset for neat plotting
# keep only first 4 groups of category
small_df <- df %>%
  filter(!is.na(.data[[num_var]]), !is.na(.data[[cat_var]])) %>%
  group_by(.data[[cat_var]]) %>%
  slice_head(n = 20) %>% # 20 observations per group
  ungroup() %>%
  slice_head(n = 80) # max 80 rows total
cat("\nRows used for boxplot:", nrow(small_df), "\n")

# 3. HISTOGRAM (numeric variable)
ggplot(small_df, aes(x = .data[[num_var]])) +
  geom_histogram(bins = 20, color = "black", fill = "skyblue") +
  labs(title = paste("Histogram of", num_var),
       x = num_var,
       y = "Frequency")

# 4. BOX PLOT (clean & clear like your example)
ggplot(small_df, aes(x = .data[[cat_var]], y = .data[[num_var]]) +
```

S101

RAJDEEPP M PARAB

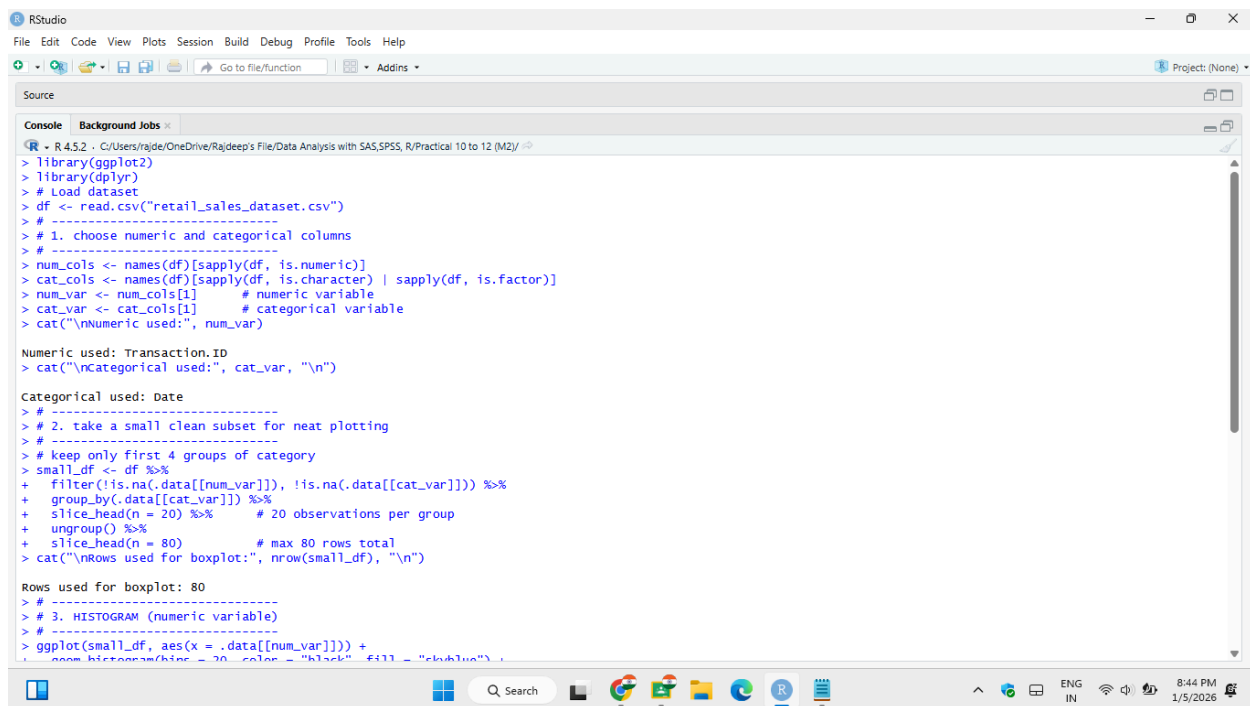
SY.BSc.CS

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```
geom_boxplot(fill = "lightblue", color = "black", outlier.color = "red") +  
labs(  
  title = paste("Boxplot of", num_var, "by", cat_var),  
  x = cat_var,  
  y = num_var  
) +  
theme_minimal() +  
theme(  
  plot.title = element_text(hjust = 0.5, size = 14, face = "bold"),  
  axis.text.x = element_text(angle = 15)  
)
```

Output:



```
RStudio  
File Edit Code View Plots Session Build Debug Profile Tools Help  
Go to file/function Addins Project: (None)  
Source  
Console Background Jobs  
R - R 4.5.2 - C:/Users/rajde/OneDrive/Rajdeep's File/Data Analysis with SAS,SPSS, R/Practical 10 to 12 (M2)/  
> library(ggplot2)  
> library(dplyr)  
> # Load dataset  
> df <- read.csv("retail_sales_dataset.csv")  
> #  
> # 1. choose numeric and categorical columns  
> #  
> num_cols <- names(df)[sapply(df, is.numeric)]  
> cat_cols <- names(df)[sapply(df, is.character) | sapply(df, is.factor)]  
> num_var <- num_cols[1] # numeric variable  
> cat_var <- cat_cols[1] # categorical variable  
> cat("\nNumeric used:", num_var)  
  
Numeric used: Transaction.ID  
> cat("\nCategorical used:", cat_var, "\n")  
  
Categorical used: Date  
> #  
> # 2. take a small clean subset for neat plotting  
> #  
> # keep only first 4 groups of category  
> small_df <- df %>%  
+ filter(!is.na(.data[[num_var]]), !is.na(.data[[cat_var]])) %>%  
+ group_by(.data[[cat_var]]) %>%  
+ slice_head(n = 20) %>% # 20 observations per group  
+ ungroup() %>%  
+ slice_head(n = 80) # max 80 rows total  
> cat("\nRows used for boxplot:", nrow(small_df), "\n")  
  
Rows used for boxplot: 80  
> #  
> # 3. HISTOGRAM (numeric variable)  
> #  
> ggplot(small_df, aes(x = .data[[num_var]])) +  
+ geom_histogram(bins = 20, color = "black", fill = "lightblue") +
```

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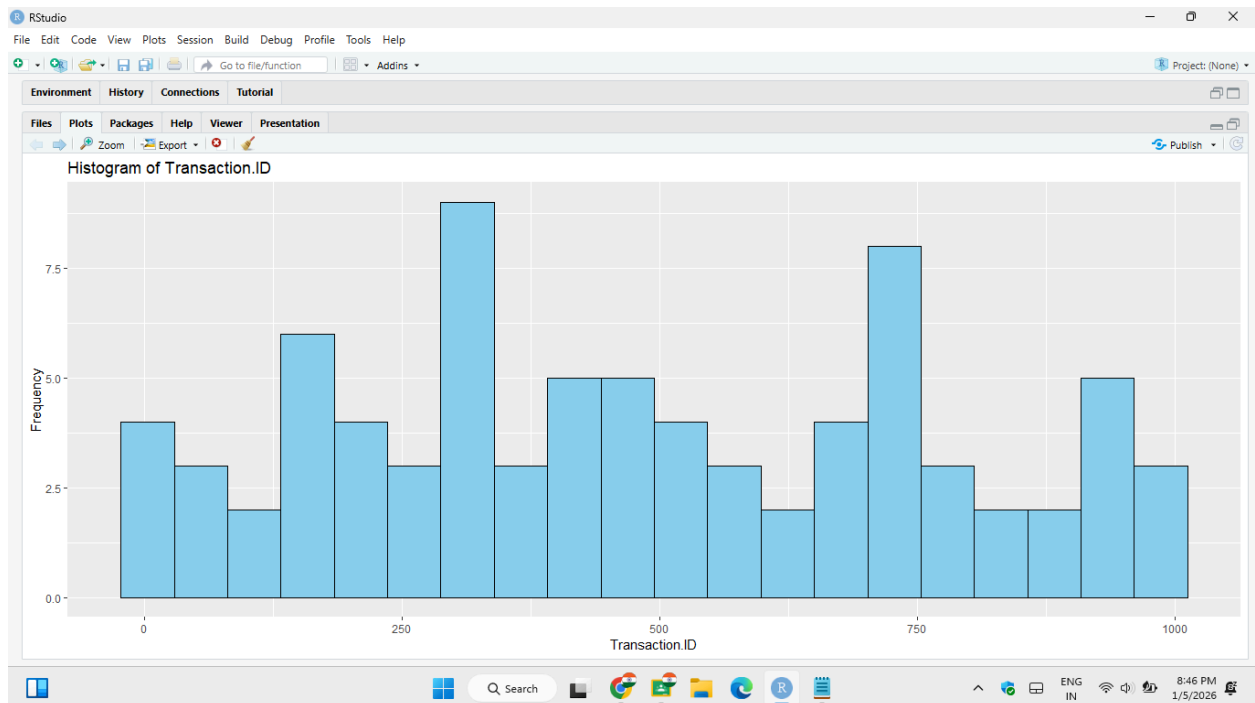
```
RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Go to file/function Addins Project: (None)

Source
Console Background Jobs
R 4.5.2 - C:/Users/rajde/OneDrive/Rajdeep's File/Data Analysis with SAS,SPSS, R/Practical 10 to 12 (M2)/

Rows used for boxplot: 80
> #
> # 3. HISTOGRAM (numeric variable)
> #
> ggplot(small_df, aes(x = .data[num_var])) +
+   geom_histogram(bins = 20, color = "black", fill = "skyblue") +
+   labs(title = paste("Histogram of", num_var),
+        x = num_var,
+        y = "Frequency")
> #
> # 4. BOX PLOT (clean & clear like your example)
> #
> ggplot(small_df, aes(x = .data[cat_var], y = .data[num_var])) +
+   geom_boxplot(fill = "lightblue", color = "black", outlier.color = "red") +
+   labs(
+     title = paste("boxplot of", num_var, "by", cat_var),
+     x = cat_var,
+     y = num_var,
+   ) +
+   theme_minimal() +
+   theme(
+     plot.title = element_text(hjust = 0.5, size = 14, face = "bold"),
+     axis.text.x = element_text(angle = 15)
+   )
>
> source("C:/Users/rajde/OneDrive/Rajdeep's File/Data Analysis with SAS,SPSS, R/Practical 10 to 12 (M2)/Practical 11 (M2).R")

Numeric used: Transaction.ID
Categorical used: Date

Rows used for boxplot: 80
> |
```



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